

Chapter 10: Independence and Conditional Probabilities

The Practice of Statistics in the Life Sciences
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Part II: Probability and Distributions

Introduction

In Chapter 9, we often studied one random phenomenon at a time. In practice, biological and scientific events are usually connected.

- Does a particular gene alter the probability of developing a disease?
- If a diagnostic test is positive, what is the probability the disease is actually present?
- Does knowing one event occurred change the probability of another?

Chapter Goal

Extend our probability toolkit from single events to **intersections, dependence, conditioning, sequential events and reverse probabilities**.

Chapter Roadmap: Four Questions to Keep Separate

1. **Can two events happen together?**

Intersections, disjoint events, and Venn diagrams.

2. **Does knowing one event occurred change the probability of another?**

Independence and dependence.

3. **What is the probability after new information is given?**

Conditional probability.

4. **How do we reverse a conditional probability?**

Tree diagrams, the Law of Total Probability, and Bayes's Theorem.

Sample Spaces and Events

A **sample space** is the set of all possible outcomes. An **event** is a set of outcomes within that sample space.

Example: Roll a fair six-sided die

$$S = \{1, 2, 3, 4, 5, 6\}$$

Let A be the event “the result is even” and let B be “the result is greater than 3”.

$$A = \{2, 4, 6\} \quad B = \{4, 5, 6\}$$

Because the six outcomes are equally likely, the probability of an event is

$$P(\text{event}) = \frac{\text{number of outcomes in the event}}{\text{number of outcomes in } S}$$

For example $P(A) = \frac{3}{6} = \frac{1}{2}$.

Combining Events as Sets

Keep $S = \{1, 2, 3, 4, 5, 6\}$, $A = \{2, 4, 6\}$ and $B = \{4, 5, 6\}$

Operation	Outcomes	Probability
$A \cap B$: even and greater than 3	$\{4, 6\}$	$\frac{2}{6}$
$A \cup B$: even or greater than 3	$\{2, 4, 5, 6\}$	$\frac{4}{6}$
A^c : not even	$\{1, 3, 5\}$	$\frac{3}{6}$

How to read “or”

The union $A \cup B$ includes outcomes in A , in B or in both. We count 4 and 6 only once.

Since $A \cap B = \{4, 6\}$, the events overlap: they are not disjoint.

Probability Notation

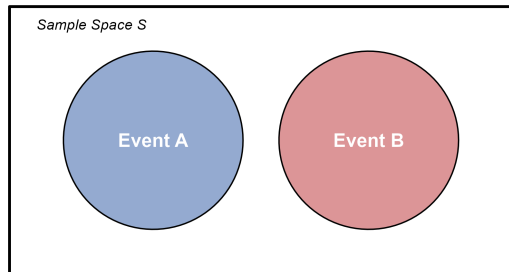
Language	Notation	Meaning
A and B	$A \cap B$	Both events occur
A or B	$A \cup B$	At least one occurs; “or” is inclusive
not A	A^c	Event A does not occur
B given A	$P(B \mid A)$	Probability of B after restricting attention to cases where A occurred

Before Calculating

Translate the question into **and**, **or**, **not** or **given**.

Visualizing Multiple Events: Disjoint Events

Disjoint Events (Mutually Exclusive)



Disjoint Events

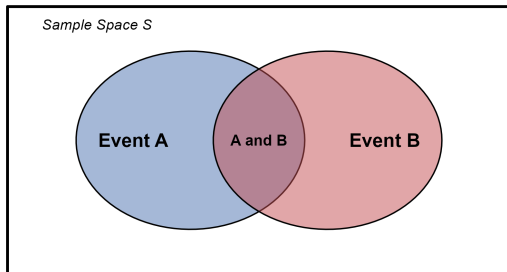
Two events are **disjoint** if they cannot occur together.

$$P(A \cap B) = 0$$

The circles do not overlap, so there is no outcome belonging to both events.

Visualizing Multiple Events: Overlapping Events

Events That Are NOT Disjoint



Most interesting scientific events are **not disjoint**.

- A patient can have high blood pressure and high cholesterol.
- The overlap represents $A \cap B$.
- A nonempty overlap can have positive probability.

Example 10.1: Two Births

Assume

- Each birth has probability 0.5 of being a girl;
- The two birth outcomes are modelled as independent.

Question

Under the simplified model, what is the probability that the next two births are both girls?

Define

A = first baby is a girl B = second baby is a girl

Translate the Question

“Both births are girls” means the intersection $A \cap B$.

Example 10.1: Two Births

	Second is B	Second is B^c
First is A	$A \cap B$	$A \cap B^c$
First is A^c	$A^c \cap B$	$A^c \cap B^c$

Under the simplified independent model, the second birth still has probability 0.5 of being a girl after we know the first birth was a girl.

Why Multiply?

To reach the “girl then girl” cell, the first event must occur **and then** the second event must occur. Because the events are independent, we multiply their probabilities.

$$P(A \cap B) = P(A)P(B) = (0.5)(0.5) = \mathbf{0.25}$$

So the model assigns a **25% probability** to two girls in two births.

The Gambler's Fallacy and Independence

Independence

Events are independent when knowing the outcome of one does not change the probability of the other.

Gambler's Fallacy

The gambler's fallacy is the mistaken belief that an outcome becomes more likely because it has occurred less often than expected. For example, if a fair die has not shown a 4 in several rolls, someone might think a 4 is “due.” Each roll is independent, however, so the probability of rolling a 4 next remains $\frac{1}{6}$.

Long-run proportions may stabilize but future outcomes do not need to compensate for a past streak.

Independent or Dependent?

- Two coin tosses
- Two cards drawn without replacement
- Taking the same cognitive test twice
- Repeated random measurement errors

Define the Events Carefully

Independence applies to the **events or random quantities you define**. Similar measurements are not automatically dependent and separate measurements are not automatically independent.

Multiplication Rule for Independent Events

Two Events

If A and B are independent

$$P(A \cap B) = P(A)P(B)$$

Three Events

If A , B , and C are **mutually independent**

$$P(A \cap B \cap C) = P(A)P(B)P(C)$$

Important Caveat

Pairwise independence alone is **not sufficient** to guarantee the three event product rule.

Do Not Assume Independence by Default

The product rule $P(A \cap B) = P(A)P(B)$ is a shortcut with a condition attached. Use it when:

- independence is explicitly stated
- independence is justified by the data-generating process
- independence has been established mathematically

Example 10.2: Rapid HIV Testing

Question

A clinic tests 200 HIV-negative people. Suppose the false-positive probability for one HIV negative person is 0.004. What is the probability that **at least one** of the 200 tests produces a false positive? For this calculation, assume the test outcomes for different people are independent.

Look for the Complement

The opposite of “at least one false positive” is much simpler: **no false positives at all**.

Example 10.2: Rapid HIV Testing

For one HIV-negative person,

$$P(\text{correct negative}) = 1 - 0.004 = 0.996$$

Because the 200 test outcomes are assumed independent,

$$P(\text{all 200 correct negatives}) = (0.996)^{200} = 0.4486$$

Now use the complement rule:

$$P(\text{at least one false positive}) = 1 - 0.4486 = \mathbf{0.5514}$$

Interpretation

Even though a false positive is rare for any one HIV-negative person, repeating the test across many people creates many opportunities for at least one error. Under this model, the chance is about **55.1%**.

Example 10.2: Rapid HIV Testing



- For one HIV-negative person, the true-negative rate is 99.6%.
- Across 200 independent tests, the chance of at least one false positive is about 55.1%.

Disjoint vs. Independent

Disjoint

Cannot occur together:

$$P(A \cap B) = 0$$

If both have positive probability, observing A tells us that B did not occur.

Independent

Knowing one event occurred does not change the probability of the other:

$$P(A \cap B) = P(A)P(B)$$

Precise Statement

Two **positive probability** disjoint events cannot be independent. A zero probability event is the mathematical exception to the blanket statement “disjoint events are never independent.”

From Special Rules to General Rules

The familiar shortcuts have conditions:

- $P(A \cup B) = P(A) + P(B)$ only when A and B are disjoint.
- $P(A \cap B) = P(A)P(B)$ only when A and B are independent.

General Rules

We now remove those restrictions so the formulas work for **any two events**.

The General Addition Rule

If A and B overlap, adding $P(A)$ and $P(B)$ counts the intersection twice.

For Any Two Events

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

If A and B are disjoint then $P(A \cap B) = 0$ and the formula reduces to

$$P(A \cup B) = P(A) + P(B)$$

Example 10.5: Dalmatians

Let

- I = the dog has hearing impairment
- B = the dog has blue eyes.

Suppose

$$P(I) = 0.28 \quad P(B) = 0.11 \quad P(I \cap B) = 0.05$$

Question

What is the probability that a randomly selected Dalmatian is hearing-impaired **or** blue-eyed?

The word “or” means the union $I \cup B$. If we simply add $0.28 + 0.11$, dogs in the overlap are counted once as impaired and a second time as blue-eyed.

Example 10.5: Dalmatians

Use the General Addition Rule

$$P(I \cup B) = P(I) + P(B) - P(I \cap B)$$

Substitute the data

$$P(I \cup B) = 0.28 + 0.11 - 0.05 = \mathbf{0.34}$$

So 34% are in at least one of the two categories. To find the probability of **neither**, use the complement:

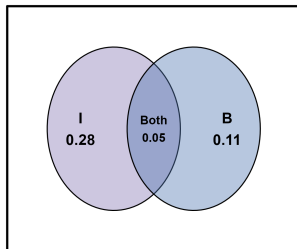
$$P(\text{neither}) = 1 - 0.34 = \mathbf{0.66}$$

Interpretation

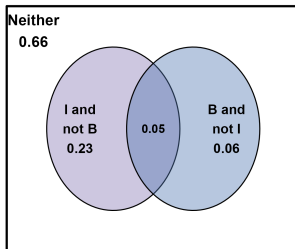
The subtraction does not remove the overlap from the final answer. It removes the **extra copy** created by adding the two marginal probabilities.

Example 10.5: Dalmatians – Visualizing the General Addition Rule

(a) Simple Probabilities of I and B



(b) Disjoint Combinations (Sum to 1)



The second diagram partitions the sample space into disjoint pieces:

$$0.23 + 0.05 + 0.06 + 0.66 = 1$$

Partition of an Event

Every dog with hearing impairment either has blue eyes or does not. These two groups cannot overlap.

Partition of an event

For any events A and B

$$A = (A \cap B) \cup (A \cap B^c)$$

Because the two parts are disjoint,

$$P(A) = P(A \cap B) + P(A \cap B^c)$$

In the Dalmatian example, I is hearing impairment and B is blue eyes. The Venn diagram gives

$$P(I) = P(I \cap B) + P(I \cap B^c) = 0.05 + 0.23 = 0.28$$

The 0.23 represents dogs with hearing impairment **and without** blue eyes.

Conditional Probability

Conditional probability answers a question after we are given new information.

Formula

When $P(A) > 0$

$$P(B | A) = \frac{P(A \cap B)}{P(A)}$$

- A is the **condition**: the information after the bar.
- B is the event whose probability we want.
- Conditioning on A makes A our **new denominator or restricted sample space**.

Example 10.3: Forest Data

Ecologists classify trees by species and life stage. The entries below are proportions of the **entire forest**.

	Dead	Live	Sapling	Total
Western red cedar (WRC)	0.02	0.10	0.08	0.20
Douglas fir (DF)	0.16	0.16	0.00	0.32
Western hemlock (WH)	0.23	0.21	0.04	0.48
Total	0.41	0.47	0.12	1.00

Question

A randomly selected tree is known to be a **sapling**. What is the probability that it is a western hemlock?

Example 10.3: Forest Dat

Before using the new information, a randomly selected tree has

$$P(WH) = 0.48$$

This 0.48 uses the **entire forest** as the denominator

What Changes?

The word “sapling” is the condition. Once we know the selected tree is a sapling, the whole forest is no longer the relevant denominator

We now restrict attention to the sapling column, which contains 12% of the forest. The question becomes a **within sapling proportion**

Example 10.3: Forest Data

The condition “sapling” restricts our attention to the sapling column.

- Saplings make up 0.12 of the entire forest.
- Western-hemlock saplings make up 0.04 of the entire forest.

Therefore, among the saplings

$$P(WH \mid \text{sapling}) = \frac{P(WH \cap \text{sapling})}{P(\text{sapling})} = \frac{0.04}{0.12} = \mathbf{0.3333}$$

Interpretation

About **33.3% of saplings** are western hemlocks. This is different from saying that 48% of all trees are western hemlocks.

Denominator Check

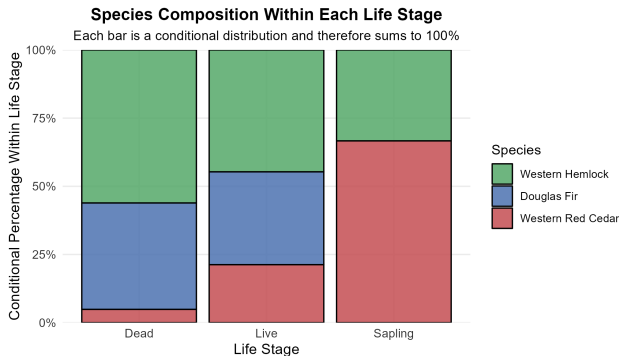
In $P(B \mid A)$, the condition A tells you which group becomes the new denominator.

Example 10.3: Forest Data

Let D , L and S mean dead, live and sapling. Let WRC, DF and WH denote the three species.

	Given dead (D)	Given live (L)	Given sapling (S)
Western red cedar	$P(\text{WRC} \mid D) = 0.049$	$P(\text{WRC} \mid L) = 0.213$	$P(\text{WRC} \mid S) = 0.667$
Douglas fir	$P(\text{DF} \mid D) = 0.390$	$P(\text{DF} \mid L) = 0.340$	$P(\text{DF} \mid S) = 0$
Western hemlock	$P(\text{WH} \mid D) = 0.561$	$P(\text{WH} \mid L) = 0.447$	$P(\text{WH} \mid S) = 0.333$
Column total	1.000	1.000	1.000

Example 10.3: Forest Data



Conditioning changes the group we are looking at.

- Entire forest: western hemlock = 48%.
- Sapling column only: western hemlock = 33%.

Key Idea

Conditional probability is a **within group proportion**.

Example 10.4: Reversing the Condition

Question

Among **western hemlock trees**, what proportion are saplings?

Previously we asked the reverse conditional probability:

$$P(WH \mid \text{sapling}) = 0.3333$$

which means: *among saplings, what proportion are western hemlocks?*

The condition is now WH , so the western hemlock row becomes the denominator:

$$P(\text{sapling} \mid WH) = \frac{0.04}{0.48} = \mathbf{0.0833}$$

Why the Answers Differ

The numerator is the same intersection, 0.04, but the denominators are different groups: 0.12 versus 0.48.

Example 10.4: Reversing the Condition

$$P(WH \mid \text{sapling}) = 0.3333$$

$$P(\text{sapling} \mid WH) = 0.0833$$

$$P(WH \cap \text{sapling}) = 0.04$$

- $P(WH \mid \text{sapling})$: among **saplings**, the percentage that are WH.
- $P(\text{sapling} \mid WH)$: among **WH trees**, the percentage that are saplings.
- $P(WH \cap \text{sapling})$: among the **entire forest**, the percentage that are both.

Conditional Probability Habit

Read the vertical bar as “given”. Say the probability statement aloud before doing the calculation.

Reversing Conditional Probabilities

A small value of

$$P(\text{evidence} \mid \text{natural causes})$$

is not necessarily a small value of

$$P(\text{natural causes} \mid \text{evidence})$$

The prosecutor's fallacy

The prosecutor's fallacy occurs when someone treats the probability of seeing the evidence *if one explanation is true* as the probability that the explanation is true *given the evidence*. For example, "This evidence would be rare under natural causes" does not establish that natural causes are unlikely.

To assess the explanation after seeing the evidence, we also need to consider how likely the other explanations were and how often they produce the same evidence. Bayes's Theorem gives us a way to do this.

The General Multiplication Rule

Rearranging the conditional probability formula gives a rule valid whether or not events are independent.

For Any Two Events

$$P(A \cap B) = P(A)P(B | A)$$

Equivalently

$$P(A \cap B) = P(B)P(A | B)$$

Example 10.6: Soda Consumption and Weight

In a survey, suppose

$$P(\text{no soda}) = 0.52$$

and

$$P(\text{overweight} \mid \text{no soda}) = 0.39$$

The 39% is not 39% of the whole population. It is 39% **within the 52% subgroup** who report no soda.

Question

What proportion of **all adults** are both non-soda drinkers and self-described as overweight?

Therefore, we need the probability of travelling through both stages: first “no soda”, then “overweight within that group”.

Example 10.6: Soda Consumption and Weight — Multiply the Stages

Apply the General Multiplication Rule:

$$P(A \cap B) = P(A)P(B | A)$$

Here

$$P(\text{no soda} \cap \text{overweight}) = (0.52)(0.39) = \mathbf{0.2028}$$

Interpretation

Approximately **20.3% of all surveyed adults** are in both groups under these survey proportions.

Association Is Not Causation

This calculation describes a joint proportion. It does not show that avoiding soda causes weight status or that weight status causes soda consumption.

How to Mathematically Test Independence

Product Definition

Events A and B are independent exactly when

$$P(A \cap B) = P(A)P(B)$$

When $P(A) > 0$, an equivalent test is

$$P(B \mid A) = P(B)$$

Similarly, when $P(B) > 0$

$$P(A \mid B) = P(A)$$

Interpretation

Independence means that conditioning does **not change** the probability.

Example 10.9: Dalmatians

Question

Are hearing impairment and blue eyes independent in the Dalmatian data?

From the earlier Dalmatian data,

$$P(I) = 0.28, \quad P(B) = 0.11$$

Independence Test

If hearing impairment and blue eyes were independent, the expected joint probability would be

$$P(I)P(B) = (0.28)(0.11) = \mathbf{0.0308}$$

This 0.0308 is the overlap we would expect **if knowing one trait gave us no information about the other.**

Applications of Conditional Probability

The same calculation appears in many fields:

- **Chemistry:** probability an alloy passes a stress test given that it contains titanium.
- **Insurance:** probability a claim came from a younger driver, and claim rates within age groups.
- **Psychology:** probability of one clinical condition given a reported symptom.

Common Structure

Conditioning means we restrict attention to a specified subgroup and compute a proportion **within that group**.

Chemistry Example

A materials laboratory tests 500 alloys.

	Pass	Fail	Total
Contains titanium	120	30	150
No titanium	70	280	350
Total	190	310	500

Question

If an alloy is known to contain titanium, what is the probability that it passes the stress test?

Chemistry Example

The condition is “contains titanium” so we use only the **150 titanium alloys** as the denominator. Among the 150 titanium alloys, 120 pass:

$$P(\text{Pass} \mid Ti) = \frac{120}{150} = \mathbf{0.80}$$

For comparison, the overall pass rate is

$$P(\text{Pass}) = \frac{190}{500} = \mathbf{0.38}$$

Interpretation

The pass rate within the titanium group is 80%, compared with 38% overall. This is strong evidence of an association between titanium status and stress-test performance in these data.

Insurance Example

Suppose

$$P(U) = 0.20 \quad P(U \cap C) = 0.08 \quad P(C) = 0.16$$

where U means “under 25” and C means “filed a claim”. These statements mean:

- 20% of customers are under 25;
- 8% of all customers are both under 25 and claimants;
- 16% of all customers filed a claim.

Questions

1. Among customers who filed a claim, what proportion are under 25?
2. Among under-25 customers, what proportion filed a claim?

Insurance Example

The condition is “filed a claim”, so claimants form the new denominator:

$$P(U | C) = \frac{P(U \cap C)}{P(C)} = \frac{0.08}{0.16} = \mathbf{0.50}$$

Interpretation

Although under-25 customers are 20% of all customers, they make up 50% of the claimants in this hypothetical portfolio.

Insurance Example

Among under 25 customers

$$P(C | U) = \frac{0.08}{0.20} = \mathbf{0.40}$$

For customers age 25 or older,

$$P(C | U^c) = \frac{0.16 - 0.08}{1 - 0.20} = \frac{0.08}{0.80} = \mathbf{0.10}$$

Interpretation

In this hypothetical dataset, the observed claim rate is 40% for under 25 customers and 10% for customers 25 or older.

Actual insurance pricing uses many variables, actuarial models and regulatory requirements; age alone does not determine a premium.

Psychology Example: Question and Given Information

Suppose a survey reports

$$P(\text{Insomnia}) = 0.25$$

and

$$P(\text{Depression} \cap \text{Insomnia}) = 0.15$$

Question

Among people who report chronic insomnia, what proportion also satisfy the study's depression criterion?

Psychology Example: Calculate and Interpret Carefully

$$P(\text{Depression} \mid \text{Insomnia}) = \frac{P(\text{Depression} \cap \text{Insomnia})}{P(\text{Insomnia})} = \frac{0.15}{0.25} = \mathbf{0.60}$$

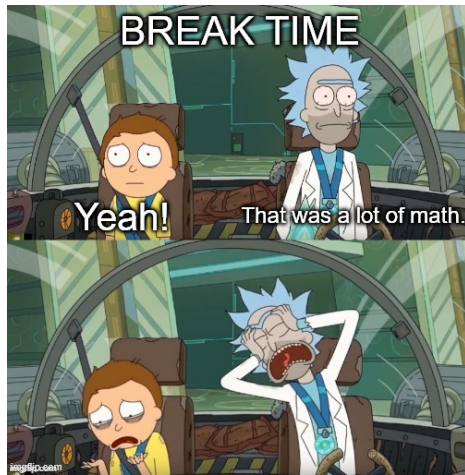
Interpretation

In this hypothetical survey population, 60% of people in the insomnia group also meet the stated depression criterion.

What This Does Not Mean

It is not an individual diagnosis and it does not show that insomnia causes depression or that depression causes insomnia.

Break Time



Tree Diagrams: Three Rules

Tree diagrams are useful when probabilities occur in **sequential stages**.

1. **Branches leaving the same node sum to 1.**
2. **Multiply along a path** to obtain the joint probability of all events on that path.
3. **Add separate disjoint paths** when several paths lead to the event of interest.

Why Trees Work

They combine conditional probability, the General Multiplication Rule, and the Addition Rule in one picture.

Example 10.10: Skin Cancer — Question and Given Information

Question

What is the overall probability that a randomly selected patient in this study is a woman?

We select a patient from a study population of people with skin cancer. **Stage 1: location of the cancer**

$$P(H) = 0.15, \quad P(T) = 0.41, \quad P(L) = 0.44$$

These probabilities sum to 1 because every patient belongs to one of these location categories.

Stage 2: sex given the location

$$P(M | H) = 0.44, \quad P(M | T) = 0.63, \quad P(M | L) = 0.20$$

The complementary woman probabilities are 0.56, 0.37, and 0.80.

Example 10.10: Skin Cancer — Plan the Solution

There are **three different ways** to end at “woman”:

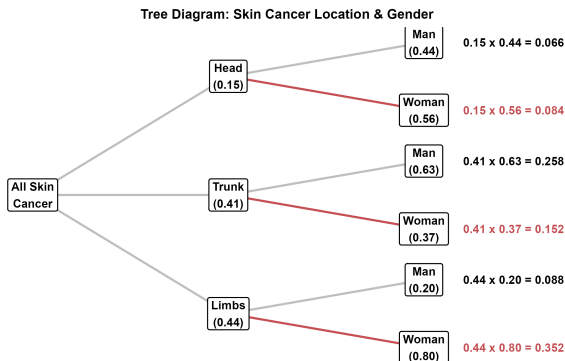
1. head and woman;
2. trunk and woman;
3. limbs and woman.

These paths are disjoint because one patient cannot be in two location categories at once.

Tree-Diagram Plan

Multiply along each complete path to get a joint probability, then add the three disjoint woman paths.

Tree Diagram: Skin Cancer — Multiply Along a Path



Consider the path “limbs, then woman”.

$$\begin{aligned} P(L \cap W) &= P(L)P(W | L) \\ &= (0.44)(0.80) = \mathbf{0.352} \end{aligned}$$

Meaning

About 35.2% of all patients are both women and in the limbs location category.

Tree Diagram: Skin Cancer — Add the Woman Paths

The three woman paths have probabilities

$$P(H \cap W) = 0.15(0.56) = 0.084,$$

$$P(T \cap W) = 0.41(0.37) = 0.1517,$$

$$P(L \cap W) = 0.44(0.80) = 0.352$$

Because these paths are disjoint,

$$P(W) = 0.084 + 0.1517 + 0.352 = \mathbf{0.5877}$$

Interpretation

Under the study proportions, approximately **58.8%** of the skin-cancer patients are women.

Example 10.11: Reverse Conditional Probability — Question First

Question

Among **women**, what proportion have their cancer in the limbs category?

We were given

$$P(W | L) = 0.80,$$

meaning 80% of patients in the limbs category are women. This reverses the direction of the conditional probability. The quantity we want is

$$P(L | W),$$

so the group of women—not the group of limbs cases—must be the denominator.

Example 10.11: Reverse Conditional Probability — Calculate

From the tree,

$$P(L \cap W) = 0.352, \quad P(W) = 0.5877$$

Therefore,

$$P(L | W) = \frac{P(L \cap W)}{P(W)} = \frac{0.352}{0.5877} = \mathbf{0.599}$$

Interpretation

Approximately 59.9% of women in the study have cancer in the limbs category.

Compare the Directions

$P(W | L) = 0.80$ and $P(L | W) = 0.599$ are different because they condition on different groups.

Why Bayes's Theorem Is Needed

Often we know a probability in one direction but need the reverse.

$$P(\text{Positive} \mid \text{Disease}) \quad \text{versus} \quad P(\text{Disease} \mid \text{Positive})$$

These are not the same.

Diagnostic-Test Language

- **Sensitivity:** $P(\text{Test}^+ \mid \text{Disease})$.
- **Specificity:** $P(\text{Test}^- \mid \text{No Disease})$.
- **Positive Predictive Value:** $P(\text{Disease} \mid \text{Test}^+)$.

Mammography Example: Question and Given Information

Question

A woman in her 60s receives a positive mammogram. Under this model, what is the probability that she actually has breast cancer, $P(C \mid +)$?

Suppose the model uses

$$P(C) = 0.0365,$$

where C means breast cancer. The test characteristics are

$$P(+ \mid C) = 0.85 \quad (\text{sensitivity}),$$

$$P(- \mid C^c) = 0.95 \quad (\text{specificity})$$

Therefore,

$$P(+ \mid C^c) = 1 - 0.95 = 0.05$$

Mammography Example: Why Sensitivity Is Not the Answer

Sensitivity tells us

$$P(+ | C) = 0.85$$

But the patient has already observed a **positive test**, so the desired probability is

$$P(C | +)$$

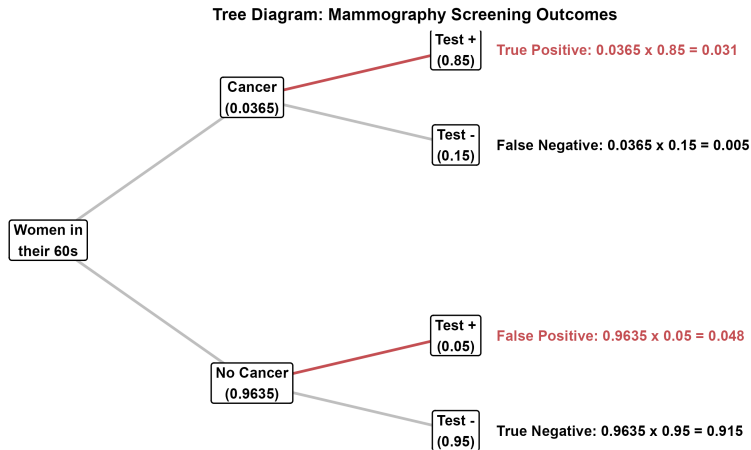
The Direction Has Reversed

“Positive given cancer” and “cancer given positive” are not interchangeable.

To answer the patient's question, we must compare two sources of positive tests:

1. true positives from the cancer group;
2. false positives from the much larger no-cancer group.

Tree Diagram: Mammography Screening



Mammography: Find the Two Positive-Test Paths

True-positive path: cancer, then positive test

$$P(C \cap +) = P(C)P(+ \mid C) = (0.0365)(0.85) = 0.031025$$

False-positive path: no cancer, then positive test

$$P(C^c \cap +) = P(C^c)P(+ \mid C^c) = (0.9635)(0.05) = 0.048175$$

Important Observation

The false-positive rate is only 5%, but it is applied to a very large no-cancer group. Consequently, the false-positive path is actually larger than the true-positive path.

Mammography: Positive Predictive Value

All positive tests come from one of the two positive-test paths:

$$P(+) = 0.031025 + 0.048175 = 0.0792$$

Of these positive tests, the cancer-and-positive branch contributes 0.031025. Therefore,

$$P(C | +) = \frac{0.031025}{0.0792} = \mathbf{0.3917}$$

Interpretation

Under this model, about **39.2% of positive screening results** occur in women who have cancer. A positive screening test is therefore not the same as a confirmed diagnosis.

Mammography with 1000 Women: Build the Groups

Natural frequencies can make the same Bayes calculation easier to understand Imagine 1000 women with the same model proportions. Because 3.65% of 1000 is not an integer, these are **expected counts** under the model

- Expected cancer cases: $1000(0.0365) = 36.5$
- Expected no-cancer cases: $1000 - 36.5 = 963.5$

Apply the test characteristics:

- Expected true positives: $36.5(0.85) = 31.025$
- Expected false positives: $963.5(0.05) = 48.175$

Why This Helps

We can compare the expected numbers of true-positive and false-positive results directly instead of thinking only in decimals

Mammography with 1000 Women: Among the Positives

The expected number of positive results is

$$31.025 + 48.175 = 79.2$$

Of these, 31.025 expected positive results come from the cancer group

$$P(C \mid +) = \frac{31.025}{79.2} = \mathbf{0.3917}$$

Base-Rate Effect

The condition is relatively uncommon. Because the no-cancer group is much larger, even a 5% false-positive rate produces more false positives than true positives

This expected-frequency calculation is the same logic as Bayes's Theorem; only the representation has changed

The Law of Total Probability

Suppose $A_1, \dots, A_k$ form a partition of the sample space: they are disjoint and exhaustive.

Total Probability

$$P(B) = \sum_{j=1}^k P(B \mid A_j)P(A_j)$$

For mammography,

$$P(+) = P(+ \mid C)P(C) + P(+ \mid C^c)P(C^c)$$

This denominator represents **all possible paths that lead to a positive test**.

The Formal Bayes's Theorem

If $A_1, \dots, A_k$ form a partition and $P(B) > 0$, then

Bayes's Theorem

$$P(A_i | B) = \frac{P(B | A_i)P(A_i)}{\sum_{j=1}^k P(B | A_j)P(A_j)}$$

- Numerator: the particular path A_i followed by B .
- Denominator: all disjoint paths that lead to B .

Practical Advice

A tree diagram often reveals the same calculation more transparently than memorizing the formula.

Example 10.14: Mammography — Question Before the Formula

Question

Given a positive mammogram, what is the probability of breast cancer, $P(C \mid +)$?

For two underlying groups, cancer and no cancer,

$$P(C \mid +) = \frac{P(+ \mid C)P(C)}{P(+ \mid C)P(C) + P(+ \mid C^c)P(C^c)}$$

- The **numerator** is the probability of the cancer-and-positive path.
- The **denominator** is the probability of **all ways to obtain a positive test**.

Same Logic as the Tree

Bayes's formula is not a new kind of reasoning. It packages the tree-diagram calculation into one equation.

Example 10.14: Mammography — Substitute the Values

Substitute the model probabilities:

$$P(C \mid +) = \frac{(0.85)(0.0365)}{(0.85)(0.0365) + (0.05)(0.9635)}$$

The numerator is

$$(0.85)(0.0365) = 0.031025,$$

while the denominator is

$$0.031025 + 0.048175 = 0.0792$$

Therefore,

$$P(C \mid +) = \mathbf{0.3917}$$

Keep the Question in View

We are asking: among all positive tests, what fraction came from the cancer branch?

The Base-Rate Effect

Bayes's Theorem combines two kinds of information:

- the **base rate or prior probability**; and
- how strongly the observed evidence is associated with each underlying state.

When the condition is rare, even a fairly good test can have a low positive predictive value because the much larger negative population can produce many false positives.

Next Examples

Lyme disease, environmental lead screening, fraud detection, and polygraph classification all use the same mathematical structure.

Bayes Example: Lyme Disease — Question and Given Information

Question

A person receives a positive Lyme-disease test. Under this model, what is the probability that the person actually has Lyme disease?

Consider a hypothetical region where

$$P(L) = 0.02,$$

so 2% of the population has Lyme disease. The test has

$$P(+ | L) = 0.90$$

and

$$P(+ | L^c) = 0.20$$

The low 2% base rate is important because 98% of the population belongs to the no-Lyme group.

Bayes Example: Lyme Disease — Compare Positive Paths

True-positive path:

$$P(L \cap +) = 0.02(0.90) = 0.018$$

False-positive path:

$$P(L^c \cap +) = 0.98(0.20) = 0.196$$

Therefore,

$$P(L \mid +) = \frac{0.018}{0.018 + 0.196} = \mathbf{0.084}$$

Interpretation

Under these hypothetical assumptions, only about 8.4% of positive results come from people with Lyme disease. The rare base rate and high false-positive rate dominate the calculation.

Bayes Example: Lead Screening — Question and Given Information

Question

A water sample turns red in the screening test. Under this model, what is the probability that the sample actually contains dangerous lead levels?

Suppose 5% of water samples contain dangerous lead levels:

$$P(L) = 0.05$$

A screening reagent turns red with probability

$$P(R | L) = 0.95$$

when lead is present, and

$$P(R | L^c) = 0.10$$

when the water is safe.

Bayes Example: Lead Screening — Calculate and Interpret

True-red path:

$$P(L \cap R) = 0.05(0.95) = 0.0475$$

False-red path:

$$P(L^c \cap R) = 0.95(0.10) = 0.0950$$

Hence

$$P(L \mid R) = \frac{0.0475}{0.0475 + 0.0950} = \mathbf{0.333}$$

Interpretation

Under this model, one third of red screening results correspond to contaminated samples. A red result is useful evidence, but confirmatory testing is still warranted.

Bayes Example: Fraud Detection — Question and Given Information

Question

A claim is flagged by the automated system. Under this model, what is the probability that the claim is actually fraudulent?

Suppose only 1% of submitted claims are actually fraudulent:

$$P(F) = 0.01$$

An automated system flags

$$P(\text{Flag} \mid F) = 0.85$$

of fraudulent claims and also flags

$$P(\text{Flag} \mid F^c) = 0.05$$

of legitimate claims.

Bayes Example: Fraud Detection — Calculate and Interpret

Fraud-and-flag path:

$$P(F \cap \text{Flag}) = 0.01(0.85) = 0.0085$$

Legitimate-but-flagged path:

$$P(F^c \cap \text{Flag}) = 0.99(0.05) = 0.0495$$

Thus

$$P(F \mid \text{Flag}) = \frac{0.0085}{0.0085 + 0.0495} = \mathbf{0.147}$$

Decision-Making Lesson

A flag is evidence that should trigger review, not proof of fraud. In this hypothetical model, about 85% of flagged claims are false positives.

Bayes Example: Polygraph Classification — Question and Given Information

Question

A person receives a positive deception indication. Under this model, what is the probability that the person is actually lying?

Suppose 10% of a particular suspect pool is lying about the matter being tested:

$$P(L) = 0.10$$

The classification procedure gives a positive indication with probability

$$P(+ | L) = 0.80$$

for a liar and

$$P(+ | L^c) = 0.15$$

for a truth-teller.

Bayes Example: Polygraph Classification — Calculate and Interpret

Lying-and-positive path:

$$P(L \cap +) = 0.10(0.80) = 0.08$$

Truthful-but-positive path:

$$P(L^c \cap +) = 0.90(0.15) = 0.135$$

Therefore,

$$P(L \mid +) = \frac{0.08}{0.08 + 0.135} = \mathbf{0.372}$$

Statistical Point

Under these hypothetical rates, a positive classification corresponds to a 37.2% probability of the underlying state. An imperfect classification is not conclusive evidence by itself.

Independent Trials Are Not the Same as the Markov Property

For idealized American roulette,

$$P(\text{red}) = \frac{18}{38}$$

on every spin, regardless of previous colours. This is **independence**.

Terminology

Independence should not be confused with the **Markov property**. A Markov process may depend on its current state; independent trials do not depend on previous outcomes at all.

Changing the size of a wager after earlier outcomes does not alter the underlying probability of the next independent spin.

Card Removal Example: Question and Given Information

Question

After ten known non-10-valued cards are removed from a standard deck, what is the probability that the next card has value 10?

In a standard 52-card deck, 16 cards have value 10 (10, J, Q, K), so initially

$$P(10) = \frac{16}{52} = \mathbf{0.307}$$

Suppose ten known non-10-valued cards are removed.

- The deck now contains only 42 cards.
- All 16 ten-valued cards are still present.

Why This Is Dependence

The first ten observed cards changed the composition of the remaining deck, so the probability of the next card depends on what was previously removed.

Card Removal Example: Calculate the Updated Probability

After the ten non-10-valued cards have been removed,

$$P(10 \mid 10 \text{ non-10 cards removed}) = \frac{16}{42} = \mathbf{0.381}$$

The probability increased from 30.7% to 38.1% because the denominator decreased while all 16 favourable ten-valued cards remained.

Learning Purpose

Sampling without replacement creates dependence. Conditional probability updates the chance of future draws as the composition of the remaining population changes.

This change in one card probability alone does not establish the expected advantage of a complete blackjack strategy.

Chapter 10 Synthesis: Which Probability Tool?

Question	Key Idea	Useful Rule
Can A and B occur together?	Disjointness	Check $P(A \cap B) = 0$
Chance of A or B ?	Union	$P(A \cup B) = P(A) + P(B) - P(A \cap B)$
Does knowing A change B ?	Independence	Compare $P(B A)$ with $P(B)$
Chance of B given A ?	Conditioning	$P(B A) = P(A \cap B) / P(A)$
Chance of both?	General multiplication	$P(A \cap B) = P(A)P(B A)$
Several stages?	Tree diagram	Multiply paths; add disjoint paths
Reverse a conditional?	Bayes	Use priors and all paths to the evidence

A Reliable Problem-Solving Workflow

1. **Define the events clearly.**
2. **Translate the wording:** and, or, not, or given.
3. **Write down known probabilities** before calculating.
4. **Check whether independence is stated or justified.**
5. For $P(B | A)$, identify the **condition A first.**
6. For a tree, verify that outgoing branches from each node **sum to 1.**
7. **Keep extra digits** during intermediate steps; round at the end.
8. **Interpret the answer in context;** association is not automatically causation or an individual diagnosis.

Final Conceptual Connection

Independence

Conditioning does **not** change the probability:

$$P(B | A) = P(B)$$

Conditional Probability

New information changes the sample space and may change the probability.

Bayes's Theorem

Reverses a conditional probability by combining the prior probability with **all paths that could have produced the observed evidence**.

These ideas form the conceptual backbone of Chapter 10.